Radionics Study Report

Machine Learning Modeling Agent

December 13, 2025

Executive Summary

This study examined the predictive capability of radiomic features for medical condition classification. The modeling results indicate insufficient evidence to support reliable classification performance.

Study Overview

This radionics study aimed to classify medical conditions using radiomic features extracted from medical imaging data. A machine learning pipeline was implemented with 10 radiomic features including texture, shape, and first-order statistics.

Performance Metrics

Metric	Value
Mean Accuracy	0.60 ± 0.20
Mean AUC	0.40 ± 0.20
Mean F1 Score	0.47 ± 0.27
Mean Precision	0.40 ± 0.30
Mean Recall	0.60 ± 0.20

Table 1: Model Performance Metrics

Key Findings

- The model achieved a mean cross-validation accuracy of 0.60 ± 0.20 across 5 folds
- While some individual predictions showed moderate confidence (scores up to 0.87), the overall performance metrics indicate limited discriminatory power
- Inconsistent performance across different validation folds
- AUC values below 0.5 in majority of folds

Decision and Justification

Decision: REJECTED

Justification: The model failed to achieve consistent predictive performance, with key metrics (AUC $<$ 0.5, F1 $<$ 0.5) indicating performance no better than random chance in most validation folds.

Conclusion

The modeling approach does not provide sufficient evidence for reliable classification of medical conditions based on the extracted radiomic features. Further feature engineering, data collection, or alternative modeling approaches may be necessary to improve predictive performance.